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Teacher Manual — Algebra 2 Honors

Florida #1200340 · Full-year Honors · Facilitator Guide

Facilitator copy. This manual contains grading guidance, rubric cut-offs, the locations of answer keys, and pacing notes. It is a teacher-only document. **Do not publish it to the student-facing course.**

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1 How this course works

Algebra 2 Honors is a full-year, asynchronous (On-Demand) course of **14 modules plus two Honors modules** (Semester 1 Honors: Probability; Semester 2 Honors: Matrices). It also runs as a live section on the same content. Roughly 500 students may be enrolled across sections, so the design keeps most practice self-scoring and reserves your time for the summative work and for outreach.

A TYPICAL LESSON

Open any lesson (for example, *Lesson 1.01 Square Roots*) and you will see the same shape:

1. **Prior Knowledge Check** — a short warm-up on the prerequisite skill, graded for completion only.
2. **Lesson content** — a short interactive lesson with worked examples and self-check practice built into the page (immediate feedback, nothing to grade).
3. **Independent Practice** — 4–10 problems that produce a graded submission. This is the main practice touchpoint.
4. **VR Lesson Resource** — a linked immersive activity for the lesson group (e.g., *Lessons 1.01–1.03 VR Lesson Resource*).

A TYPICAL WEEK (ON-DEMAND)

SEGMENT	WHAT THE STUDENT DOES	GRADED?
Start of each mini-lesson	Prior Knowledge Check	Completion
Lesson body	Interactive content + embedded self-checks	No
After each mini-lesson	Independent Practice (4–10 problems)	Yes (points)
Mid-module	Mid-Module Check(s)	Yes (points)
VR activity	Immersive lesson resource	Participation
End of module	Summative Review, then Summative Assessment	Review = completion (unlimited attempts); Summative = points

MAJOR ASSESSMENTS

Each of the 14 modules ends with a **Module Summative Assessment** preceded by an unlimited-attempt **Summative Review**. Several modules also carry one or more **Mid-Module Checks**. Modules 4, 11, and 13 include **Honors versions** of the summative or additional Honors lessons; the two Honors modules (Probability, Matrices) are Honors-only. These summatives are the graded work that actually costs your time — plan around them (see §10).

2 Ungraded interactives & module structure

Ungraded interactives. The self-checks and worked-example practice inside each lesson give students immediate feedback and *cost you nothing* — they produce no submission and need no grading. Treat them as formation-by-practice, not busywork to chase. A student who skips them will show up indirectly in weaker Independent Practice and Mid-Module Check performance, not as a missing grade.

Summative Reviews are completion items with unlimited attempts. They exist so students can rehearse the whole module before the graded summative. Encourage repeated attempts; do not treat a low first attempt as a signal of anything but effort.

> **Highest-leverage move:** Each module has a teacher-only *Lesson Maps* page and a *Teacher Resources (DO NOT PUBLISH)* section. Read the module's Lesson Map before the module opens so you know exactly which skills each Independent Practice and Mid-Module Check targets. This is where your answer keys and pacing notes live (for example, *Lesson 6.05 Independent Practice Answer Key*). Keep these unpublished.

3 AI & VR in this course (student-facing)

AI. The course's posture is AI *awareness*, not required tool use. Students may use AI to have a concept re-explained, to check understanding, or to generate extra practice — but their graded reasoning and solutions must be their own, and they briefly note when they used a tool. Speak to this the same way the syllabus does; do not add new AI requirements.

VR. Each lesson group links a **VR Lesson Resource**, and the Start Here section includes an *Accessing OptimaED Content, Locations, and IFX* page for setup. Describe VR to students by what they do in it — explore a concept immersively, then return to Canvas to complete the practice. If a student cannot get into VR, there is an alternative path so they do not fall behind. VERIFY Optima is moving VR platforms; confirm current access steps and vocabulary before pointing students at the setup page.

4 Using Ignite AI to support students (teacher-facing)

VERIFY Ignite AI's exact features vary by Canvas release and by what Optima has enabled — confirm against your instance before relying on any one button. Used well, the Canvas AI assistant helps you make a 500:1 load feel personal:

- **Draft individualized feedback** on a summative that you then edit and own — faster turnaround, same human judgment.
- **Summarize quiz analytics** to find the two or three problems the cohort missed most, then post one targeted announcement or worked example.
- **Surface students who need a nudge** (missed work, a declining pattern across mid-module checks) and draft a warm check-in you send after editing.
- **Generate practice variants** for a student who needs another rep on a specific skill.

> **Standing rule:** AI drafts; the teacher decides. You read, edit, and own anything sent to a student or family. AI is never used to surveil, rank, or auto-penalize.

5 Grading categories & gradebook setup

The exported course ships with a single assignment group (`Assignments`, unweighted), so by default it is scored on **total points**. VERIFY Confirm your section's grading policy. If your school prefers weighted categories, this is a clean starting scheme — create the groups in Canvas and set weights:

ASSIGNMENT GROUP	WHAT GOES HERE	SUGGESTED WEIGHT	METHOD
Practice & Checks	Independent Practice, Mid-Module Checks	50%	Auto / points
Module Summatives	Module Summative Assessments (Honors versions where applicable)	40%	Auto + rubric on any free response
Completion & VR	Prior Knowledge Checks, Summative Reviews, VR participation	10%	Completion

Setup steps: Gradebook → Settings to enable weighted groups → create the three groups above → set weights → mark Prior Knowledge Checks and Summative Reviews as complete/incomplete → leave late/missing policies off (On-Demand pacing is handled at the quarter boundary, per the syllabus). Use the **module requirements / progress view** as your at-a-glance completion signal across the ungraded scaffolding.

6 Rubrics for teacher-graded items

Most Independent Practice, Mid-Module Checks, and Summatives are auto-scored. Where a summative includes a **free-response / show-your-work** item, attach this consistent 4-criterion rubric as a native Canvas Rubric so it scores in SpeedGrader. Keep it identical across modules so grading stays near ~5 minutes per student.

CRITERION	FULL CREDIT	PARTIAL	NONE
Correctness of result	Correct answer, correct units/form	Minor arithmetic slip, method sound	Incorrect and method unclear
Mathematical reasoning	Valid, complete method shown	Method mostly right, a gap	No valid method shown
Clarity of work	Steps legible and ordered; notation correct	Followable with effort	Cannot follow the work
Interpretation (context items)	Answer interpreted in context; extraneous solutions addressed	Partial interpretation	No interpretation

> **Answer keys are teacher-only.** Module answer keys live in each module's *Teacher Resources (DO NOT PUBLISH)* section and Lesson Maps (e.g., *Lesson 6.05 Independent Practice Answer Key*). Use them to build the full-credit shape for any hand-scored item; never publish them to students. VERIFY Confirm which summatives contain free-response items in your section's copy.

7 Quizzes & assessment activities: how to run & grade

- **Prior Knowledge Checks** — completion. Do not grade for correctness; they diagnose readiness. A pattern of blank checks is an early-outreach signal.
- **Independent Practice** — auto-graded points. Skim the item analysis; if one skill is missed widely, post a short worked example rather than grading individually.
- **Mid-Module Checks** — auto-graded points. Use these as your mid-module triage: low scores clustered on one lesson tell you exactly what to reteach in an announcement or office hour.
- **Summative Reviews** — completion, unlimited attempts. Encourage retakes.
- **Module Summative Assessments** — the graded checkpoint. Auto-score what can be auto-scored; apply the §6 rubric to any free-response item.
- **VR Lesson Resources** — participation. Review captures/participation rather than scoring correctness.

No discussion forums. This course does not ship graded discussion boards; community happens in office hours, VR, and announcements. If you want a discussion, add it yourself and keep it low-stakes.

8 Coach, not catch

- Feedback names the next step, not just the error. Lead with what's working in the student's method.
- Practice and reviews allow re-attempts; they are learning tools, not traps.
- Use analytics and Ignite AI to **find students who need help early**, then reach out — not to build a case against them.
- On suspected AI misuse or copied work, open a conversation: ask the student to talk through their solution. Treat it as formation.
- When you post a common-error note to the class, model the correct thinking; never single out a student's weak work.

9 Weekly announcement scripts

For an asynchronous cohort, the weekly announcement is your main presence. Clone this template each week:

> *Scholars — this week you should be in **[Module/Lesson]**. Focus on **[skill]**. The one thing not to miss: **[due summative / tricky concept]**. If your last Mid-Module Check on **[topic]** felt shaky, come to office hours **[day/time]** — bring the problem that tripped you up. Keep going; this is hard and you are doing it.*

Module 1 example: "Welcome to Algebra 2 Honors. Start with Lesson 1.01 and keep Desmos open. New this module: imaginary and complex numbers — strange at first, then routine. Don't miss the Module 1 Summative at the end. Office hours [day/time]."

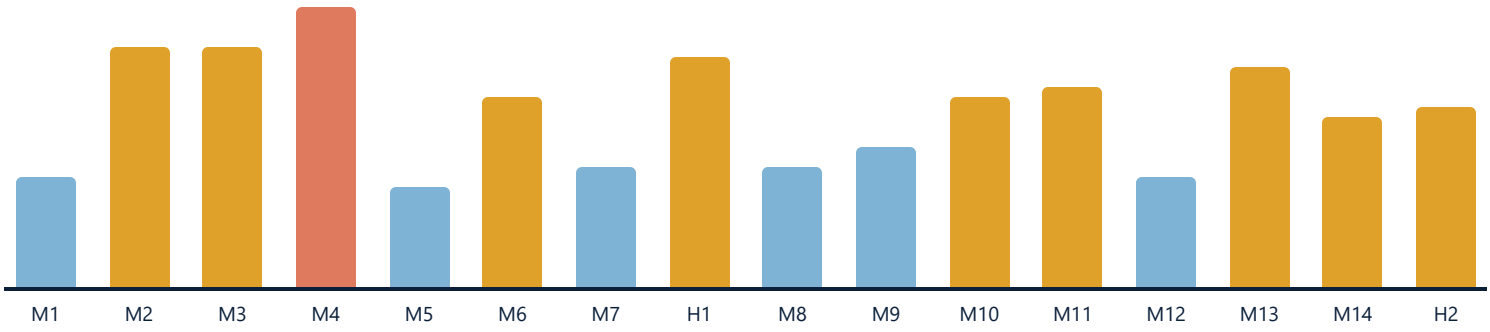
Pre-summative example: "The Module [N] Summative opens Friday. Use the Summative Review first — unlimited attempts — and note which problems you have to look up. Those are exactly what to review."

Re-engagement example: "A few of you have not submitted in over a week. Nothing is broken — just reply and tell me where you're stuck. We'll make a plan."

10 Teacher workflow & workload map

A normal week, in order: scan Mid-Module Check and Independent Practice analytics → reach out to any student with a week of missing work → grade any free-response summative items with the §6 rubric → post the weekly announcement → answer Canvas Inbox messages. Auto-scored items carry almost no grading time; your time concentrates around each module's summative.

Relative facilitator-grading load by module (driven by lesson count and Honors summatives):



light moderate heavy — plan extra time on the coral/gold modules (more lessons or an Honors summative). VERIFY heights are estimated from lesson counts, not measured grading hours.

11 Standards coverage map (full year)

Florida B.E.S.T. standards (CPALMS #1200340), mapped to the module that carries each. Read from the course's Scope & Sequence documents. **Teacher-only** — standard codes are not shown to students. **(H)** marks Honors-only benchmarks.

SEMESTER 1 — BY MODULE

MODULE	STANDARDS
M1 Square Roots & Complex Numbers	NSO.2.1
M2 Quadratic Equations in One Variable	AR.1.1, AR.1.3, AR.1.8, AR.3.2, AR.3.3
M3 Quadratic Functions & Inequalities (2-var)	AR.3.4, AR.3.8, AR.3.9, AR.3.10, F.2.2, F.2.3, F.2.5
M4 Polynomial Equations in One Variable	AR.1.3, AR.1.5, AR.1.6, AR.1.8, AR.6.1, AR.6.2, **(H)** AR.1.11
M5 Polynomial Functions in Two Variables	AR.6.5, F.2.2, F.2.3, F.2.5
M6 Rational Expressions & Functions	AR.1.9, AR.8.1, AR.8.2, AR.8.3
M7 Radical Equations & Functions	AR.7.1, AR.7.2, AR.7.3, F.2.2, F.2.3, F.2.5
Honors Probability	DP.4.1, DP.4.2, DP.4.3, DP.4.4, DP.4.9, DP.4.10

SEMESTER 2 — BY MODULE

MODULE	STANDARDS
M8 Rational Exponents & Radical Expressions	NSO.1.3, NSO.1.5
M9 Logarithms & One-Variable Equations	NSO.1.6, NSO.1.7, AR.5.2, AR.5.5, FL.3.1, FL.3.2, FL.3.4
M10 Exponential & Log Functions (2-var)	AR.5.7, AR.5.8, AR.5.9, F.2.2, F.2.3, F.2.5
M11 Properties of Functions	F.1.1, F.1.7, F.1.9, **(H)** AR.10.1, **(H)** AR.10.2
M12 Inverse Functions	F.3.2, F.3.4, F.3.6, F.3.7
M13 Absolute Value & Piecewise Functions	AR.4.2, AR.4.4, **(H)** AR.9.10
M14 Systems of Equations & Inequalities	AR.9.2, AR.9.3, AR.9.5, AR.9.7
Honors Matrices	**(H)** NSO.4.1, NSO.4.2, NSO.4.3, NSO.4.4

THREADED THROUGHOUT THE YEAR

The Mathematical Thinking & Reasoning standards **MA.K12.MTR.1.1–7.1**, the ELA expectations **ELA.K12.EE.1.1–6.1** (cite evidence, comprehend, infer, discuss, format, voice), and **ELD.K12.ELL.MA.1** are embedded across every module rather than taught as separate lessons.

VERIFY A few CPALMS benchmarks in the full #1200340 description (e.g., DP.2.8/DP.2.9 quadratic/exponential regression) are not listed as standalone lessons in the Scope & Sequence; confirm whether your section addresses them inside modeling tasks or treats them as extensions.

12 Adapting the async lessons to a live class (2 days/week)

The course is built for On-Demand study, but a live section can run on the same Canvas content by using class time for the parts students most need a teacher for — worked examples, misconceptions, and questions — and pushing the routine practice to independent work. Below is a repeatable **weekly rhythm for a class that meets twice a week** (roughly one module every 1–2 weeks; give heavy modules like M2, M3, M4 two weeks).

THE WEEKLY TWO-SESSION PATTERN

WHEN	IN THE LIVE SESSION (60–75 MIN)	ASYNC BEFORE/AFTER (IN CANVAS)
Session 1 (new material)	Open with the Prior Knowledge Check as a warm-up (2–3 problems on the board). Teach the first mini-lessons of the module live, working the same examples the online lesson uses. Do 2–3 problems together, then have students start that lesson's Independent Practice while you circulate.	*Before:* skim the lesson page so you're primed. *After:* finish the Independent Practice for the lessons covered; watch the online lesson again if needed.
Session 2 (practice & check)	Start with a 5-minute review of the most-missed Independent Practice item (pull it from analytics). Teach the remaining mini-lessons, then run the **Mid-Module Check** in class or a live practice set as a formative gauge. Use the last 10 minutes for the VR Lesson Resource or a quick exit problem.	*After:* complete any remaining Independent Practice and the **Summative Review** (unlimited attempts) before the next meeting. Take the **Module Summative** online, on your own, by the posted date.

WEEK-BY-WEEK SUGGESTIONS

- **Pacing:** One module per week for light modules (M1, M5, M7, M12); two weeks for heavy ones (M2, M3, M4, M6, M13) and the Honors modules. Aim to finish Semester 1 (M1–M7 + Probability) by the winter break.
- **Summatives stay asynchronous.** Keep the Module Summative as independent, out-of-class work so live time is spent teaching, not testing. Use Mid-Module Checks as your in-class formative pulse.
- **Let analytics drive Session 2.** The single most useful live move is to reteach the one or two problems the class missed most on the Independent Practice — you already have that data in the gradebook.
- **VR as the live hook.** The VR Lesson Resources are natural whole-class or paired activities; anchor one per module to a live session rather than leaving it optional.
- **Honors depth in class.** Use live time for the Honors-only topics (Binomial Theorem, sequences, piecewise, probability, matrices) and richer contexts — that is where Honors rigor is easiest to add live.
- **Missed a class?** Because every lesson also lives in Canvas, an absent student can self-serve the full lesson online — point them to the same page you taught from.

VERIFY Adjust session length and per-module weeks to your school's bell schedule and term calendar.

13 What makes this course classical & cultivating culture

Classical moves in this course: mathematics is taught as a body of reasoning, not a bag of tricks — students meet *why* a method works (complex numbers as a genuine extension of number, the Remainder Theorem, inverses as undoing). The MTR standards ask for perseverance, multiple representations, and justifying results, which is the classical habit of reasoned demonstration. The course moves deliberately from concrete (numerical work) to abstract (general functions), honoring how understanding actually grows.

Cultivating culture asynchronously: name students' good reasoning publicly in announcements; model curiosity ("here's where complex numbers came from"); treat mistakes as the ordinary path to fluency; keep the seven virtues visible — especially perseverance and honesty in showing work. In a live section, protect a few minutes for students to compare methods; the classical classroom prizes the discussion of *how* we know.

14 Sensitive-content notes

This is a mathematics course; it does not contain mature themes, contested texts, or sensitive imagery that would warrant advance family notice. The only student-experience note worth flagging is **VR access**: some families may have questions about headset use and time. A short, one-paragraph note at course start — what VR is used for, roughly how often, and the non-VR alternative — heads off most questions. Keep it factual and invite families to reach out.
